LAB EXPERIMENT

14. Perform Perspective Transformation on the given image using python and Open CV.

Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread('image.jpg')
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

rows, cols = img.shape[:2]

src = np.float32([
    [0, 0],
    [cols - 1, 0],
    [0, rows - 1],
    [cols - 1, rows - 1]
])

dst = np.float32([
    [50, 50],
    [cols - 50, 0],
    [0, rows - 50],
    [cols - 1, rows - 1]
])

M = cv2.getPerspectiveTransform(src, dst)
result = cv2.warpPerspective(img, M, (cols, rows))

plt.subplot(1, 2, 1)
plt.imshow(img)
plt.title('Original Image')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(result)
plt.title('Perspective Transformation')
plt.axis('off')

plt.show()
```

Original Image



Perspective Transformation

